

YVAN BOTA

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PROFESSIONAL STATEMENT

Electrical Engineering student with hands-on experience designing embedded hardware, PCB systems, and power electronics from concept through testing and integration. Experienced in circuit simulation, firmware development, and hardware validation. Research Scholar in solid-state lithium metal batteries with experience modeling BMS interfaces in MATLAB/Simulink. Passionate about hardware/software co-design for high-performance systems.

EDUCATION

MS - Electrical and Computer Engineering, Kennesaw State University - GPA: 4.0/4.0 January 2026 – May 2027

BS - Electrical Engineering, Kennesaw State University - GPA: 3.92/4.0 January 2022 – December 2025

Relevant Coursework, Power Electronics, Embedded Systems, Control Systems, FPGA Design, Digital System Design, Microprocessors, Analog IC Design, Computer Architecture

CERTIFICATES

Fundamentals of Engineering (FE) – Electrical and Computer, NCEES [☑](#) | Verilog HDL for Digital Design and Verification [☑](#)

TECHNICAL SKILLS

Hardware & PCB Design: PCB layout & fabrication, multilayer board design, power distribution systems, embedded hardware prototyping

Embedded & Hardware Systems: ESP32, PIC16F877/18F45K22, Raspberry Pi, Arduino, FPGA Development

Programming: C, C++, Python, MATLAB, Bash/Linux

Interfaces & Protocols: I²C, SPI, UART, GPIO, PWM, ADC/DAC

Tools & Platforms: MATLAB/Simulink, LabVIEW, Oscilloscopes, Hardware Debugging

Testing & Integration: System Validation, Hardware Debugging, Test Execution, Oscilloscopes, Multimeters, Power Supplies, LabVIEW

PROFESSIONAL EXPERIENCE

Griffiss Institute | Air Force Research Laboratory (AFRL), Syracuse University, June 2025 – August 2025

Undergraduate Research Intern – Summer 2025 Research Scholar Program

- Built and validated MATLAB/Simulink (Simscape) equivalent-circuit models from experimental EIS data, extracting bulk/interfacial resistance, capacitance, and diffusion parameters relevant to analog system behavior.
- Fabricated and tested 20+ sensor-driven CR2032 coin cells (liquid and solid-state), analyzing electrochemical performance and optimizing composite electrolytes for stability and reduced impedance.

TAMTEK SYSTEMS, Assistant Electronics Systems Designer Intern July 2023 – September 2023

- Designed and debugged PIC-based embedded automation systems integrating sensors, actuators, and GSM-enabled monitoring modules for fire and security applications.

PROJECTS

Smart Automatic Transfer Switch (ATS) January 2024 – June 2024

- Engineered a voltage-sensed power transfer system with threshold-based switching logic, ensuring stable load transition and demonstrating real-world power delivery control under dynamic supply conditions

Portable Emissions Tracer (Selected for NCUR Expo), August 2025 – December 2025

Inaugural Winners – Robin and Doug Shore Entrepreneurship & Innovation Award

- Benchmarked a portable air quality monitoring system with wireless data logging achieving 8-10hrs of operation.

MyoWare Prosthetic Hand with 5-DOF Control January 2026 – May 2026

Developed an EMG-controlled prosthetic hand using MyoWare muscle sensors and embedded actuation to achieve multi-degree-of-freedom finger motion for assistive robotics applications.

FPGA-Based Matrix Multiplication Accelerator (AI Hardware Project) January 2026 – May 2026

- Designed a 32×32 HLS-based accelerator optimized via pipelining and loop unrolling, analyzing throughput vs. resource (LUTs, DSPs, BRAM)/ power trade-offs to emulate energy-efficient compute architectures

LEADERSHIP & SERVICE

President, Internet of Things (IoT) Club January 2024 – January 2025

- Led multidisciplinary engineering teams developing embedded hardware, PCB systems, and IoT prototypes.

Graduate Teaching Assistant (GTA), Kennesaw State University January 2026 – Present

- Led and supported EE 2301, EE 2305, and EE 2501 laboratory sessions, reinforcing core circuits, and electronics.